Explicit Semantic Transition Modeling for Open-Vocabulary Remote Sensing Change Detection

Yikun Liu, Ying Li, Zhihao Li, Minghao Liu, and Gongping Yang

***Abstract*—Open-vocabulary change detection (OVCD) aims to detect land-cover changes associated with semantic concepts, but existing methods usually infer change semantics from independently predicted endpoint labels. Such endpoint-centric formulations do not directly model semantic change as a directed from-to transition and may suffer from accumulated endpoint errors. In this article, we propose an explicit semantic transition modeling framework for open-vocabulary remote sensing change detection. The proposed method uses frozen DINOv2 and SegEarth-OV3 to extract bi-temporal visual features, semantic probability maps, and text prototypes. A lightweight proposal branch identifies ROIs that may contain changes, within which the model directly predicts pixel-level directed semantic transitions. To handle the large transition space, a compact Top-*k* candidate set is first constructed from ROI semantic distributions, and a residual candidate supplement module (RCSM) further supplements potentially omitted transitions. Moreover, a consistency-aware semantic transition refinement module (CSTRM) calibrates transition logits through transition-marginal consistency (TMC) and improves spatial coherence through transition-aware refinement. Experiments on six remote sensing change detection datasets demonstrate that the proposed formulation consistently outperforms representative unsupervised, foundation-model-based, and open-vocabulary methods. The code will be available at https://github.com/liuyk97/OV-SCD**

***Index Terms*—Open-vocabulary learning, remote sensing change detection, semantic transition modeling, vision–language models.**

I. Introduction

REMOTE sensing change detection aims to identify land-cover changes from bi-temporal or multitemporal observations of the same geographic area and has been widely studied for urban planning, resource management, environmental monitoring, and disaster assessment [1], [2], [3], [4]. Conventional binary change detection (BCD) focuses on localizing changed regions [5], [6], [7], [8], [9], [10], while semantic change detection (SCD) further identifies the semantic categories involved in the changes [11], [12], [13], [14], [15], [16]. Despite remarkable progress, most existing change detection methods are developed under a closed-set assumption, where the categories of interest are predefined by the training datasets. This assumption limits their scalability in real-world applications, where land-cover changes may involve diverse, long-tailed, or user-specified categories. With the rapid development of vision–language models and open-vocabulary semantic segmentation, open-vocabulary change detection (OVCD) has recently emerged as a more flexible paradigm for detecting changes associated with arbitrary textual or semantic concepts [17], [18], [19], [20].

Recent OVCD studies mainly build upon foundation models by decomposing change understanding into several reusable components, such as mask proposal, temporal comparison, and semantic identification. DynamicEarth formulates representative training-free paradigms that combine proposal generators, comparators, and open-vocabulary identifiers to localize and recognize category-specific changes [20]. Seg2Change further adapts open-vocabulary semantic segmentation models to change detection by using category-agnostic change maps to bridge bi-temporal semantic segmentation and change localization [21]. Other recent methods also improve OVCD by enhancing foundation-model collaboration, semantic consistency, response calibration, or prompt-driven inference [22], [23], [24]. These studies demonstrate the feasibility of extending change detection beyond fixed-label spaces, but most of them still infer change semantics from masks, semantic responses, or separately predicted endpoints.

Although existing OVCD formulations are effective for category-specific change localization, they do not directly model semantic change as a directed from-to transition. As illustrated in Fig. 1, indirect endpoint composition first estimates the semantic endpoints of the two temporal phases and then derives the transition by combining them. This indirect formulation is vulnerable to accumulated transition errors, since an incorrect endpoint prediction at either temporal phase may lead to an incorrect transition, even when the bi-temporal visual evidence of change is clear. More importantly, land-cover evolution is inherently directional. For example, "vegetation to building" and "building to vegetation" correspond to opposite semantic processes, but endpoint composition does not explicitly learn the directed evidence that distinguishes these two transitions. Therefore, rather than com-

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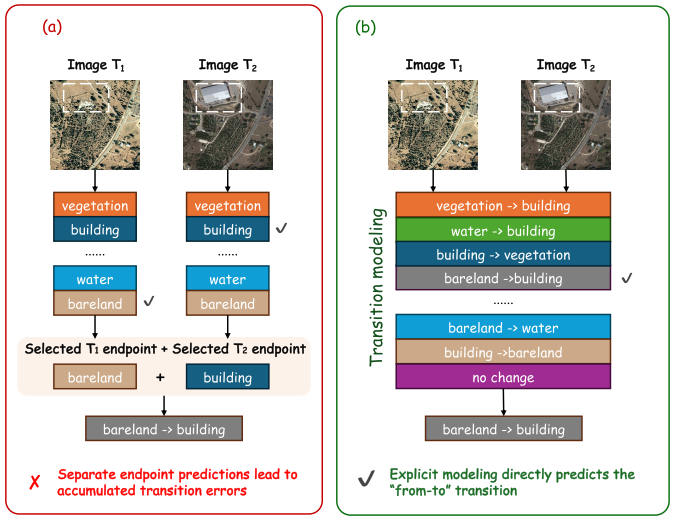


Fig. 1. Motivation of explicit semantic transition modeling. Unlike methods that compose final labels from independently predicted endpoints, our formulation directly predicts directed from-to transitions from bi-temporal images.

pletely discarding endpoint semantics, we retain the endpoint semantic distributions as candidate priors to reduce the transition search space, while the final transition is determined by explicit transition scoring based on bi-temporal observations instead of being obtained by directly composing independently predicted endpoint labels. This formulation does not eliminate classification errors; rather, it avoids requiring two independently predicted endpoint labels to be simultaneously correct and directly optimizes the transition label required by the task.

However, introducing explicit semantic transition modeling also raises two practical challenges in open-vocabulary scenarios. First, the quadratic growth of the transition space is a computational cost introduced by the explicit formulation itself, rather than a limitation of endpoint-based methods. Scoring all possible $i \rightarrow j$ transitions is inefficient and often introduces many irrelevant candidates. A small candidate transition set built only from endpoint semantic distributions is more efficient, but it may omit plausible transitions when the endpoint responses are uncertain. Thus, explicit transition prediction requires a candidate construction strategy that is both efficient and sufficiently complete. Second, transition logits should be reliable not only in terms of semantic discrimination but also in terms of transition-marginal consistency (TMC) and spatial coherence. In open-vocabulary remote sensing scenes, semantic responses are often noisy or fragmented due to scale variation, complex land-cover layouts, and cross-temporal appearance changes. Without proper refinement, pixel-level transition logits may become locally inconsistent or weakly supported by their induced from-to transition marginals. These challenges motivate a framework that directly predicts explicit transitions while improving candidate coverage and transition reliability.

To this end, we propose an explicit semantic transition modeling framework for OVCD. Here, "explicit" means that each directed from-to semantic transition, such as "vegetation to building," is treated as a prediction target rather than being obtained by composing two independently predicted endpoint labels. Given a pair of bi-temporal images and a prompt vocabulary, frozen DINOv2 [25] and SegEarth-OV3 [26] are employed to extract bi-temporal visual features, semantic probability maps, and text prototypes. The semantic probability maps provide the spatial probability of each prompted category, whereas the text prototypes provide language-aligned category embeddings. A change probability map is first generated to provide change proposals, followed by ROI pooling to obtain ROI-level visual and semantic representations. Based on the pooled ROI semantic distributions, a Top-k candidate transition set C is initialized by combining the most likely from-to endpoints. The explicit transition modeling module then selects transition prototypes from C and predicts pixel-level transition logits with the query map, producing explicit $i \rightarrow j$ transition responses without using endpoint composition as the final prediction rule. To reduce candidate omission, the residual candidate supplement module (RCSM) predicts a Top-m residual transition set and supplements it into C. To improve transition reliability, the consistency-aware semantic transition refinement module (CSTRM) further refines transition logits through TMC for semantic calibration and transition-aware spatial refinement (TSR) for spatial refinement.

In summary, the contributions of this work are as follows.

1) To formulate OVCD around directed transitions, we explicitly represent semantic changes as $i \rightarrow j$ transitions rather than using independently predicted endpoint labels as the final prediction rule.
2) To improve candidate coverage under large open-vocabulary transition spaces, we propose RCSM to supplement the Top-k candidate transition set C with Top-m residual transitions.
3) To enhance transition reliability, we propose CSTRM with TMC for semantic calibration and TSR for spatial refinement, improving TMC and regional coherence of transition logits.

II. Related Work

A. Remote Sensing Change Detection

Remote sensing change detection has been extensively studied for identifying land-cover variations from bi-temporal or multitemporal observations. Early BCD methods mainly rely on handcrafted difference modeling, including change vector analysis, image differencing, threshold segmentation, clustering, and subspace transformation-based methods [1], [27], [28], [29], [30], [31]. Although these methods are simple and annotation-efficient, their low-level discrepancy measures are sensitive to illumination variations, seasonal changes, registration noise, and sensor-induced appearance differences, often leading to pseudo changes in complex remote sensing scenes.

With the development of deep learning, BCD has shifted toward learnable bi-temporal representation learning. Fully convolutional Siamese networks introduce shared-weight encoders to extract comparable features from image pairs [2], while subsequent methods improve feature interaction and spatial detail recovery through multiscale fusion,

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attention mechanisms, densely connected architectures, and difference-enhancement strategies [3], [4], [32], [33], [34], [35]. More recently, transformer-based methods such as BIT and ChangeFormer further model long-range spatial–temporal dependencies for change detection [35], [36]. These methods have significantly improved change localization accuracy, but their outputs remain category-agnostic and cannot describe the semantic content of land-cover changes.

SCD further provides semantic interpretation by jointly localizing changed regions and identifying their categories. Existing methods usually formulate SCD as a multitask learning problem, where semantic segmentation and change detection are optimized together to exploit their mutual constraints [37], [38], [39]. The construction of high-resolution SCD benchmarks, such as HRSCD [37], SECOND [14], and Landsat-SCD [40], has also promoted pixel-level semantic change understanding.

However, most SCD methods still rely on predefined semantic labels and infer changes from separately learned temporal representations or endpoint semantic predictions. Such a closed-set formulation limits their ability to recognize unseen, long-tailed, or user-specified land-cover changes, and it does not explicitly represent semantic change as a directed from-to transition. This motivates a more flexible formulation that goes beyond fixed-label spaces and directly reasons about semantic transitions in open-vocabulary scenarios.

B. Vision–Language Models

Vision–language models have recently become an important foundation for open-vocabulary visual perception. By aligning visual and textual representations through large-scale image–text pretraining, CLIP enables zero-shot recognition of categories specified by natural language prompts [17]. Following this paradigm, a series of foundation models have further extended open-vocabulary perception to localization and segmentation tasks. For example, ALIGN improves vision–language representation learning with noisy image–text pairs [41], GLIP unifies localization and vision–language pretraining [42], Grounding DINO supports text-driven object grounding [43], and SAM provides promptable mask generation with strong zero-shot transfer ability [18]. These models provide reusable semantic and spatial priors for downstream tasks, making it possible to define categories at inference time rather than relying only on fixed training labels.

However, remote sensing imagery differs substantially from natural images in terms of overhead viewpoint, scale variation, dense object layouts, and complex land-cover patterns. Therefore, directly transferring general-purpose vision–language models to remote sensing scenes may suffer from domain gaps and inaccurate dense predictions. To address this issue, several remote sensing-oriented foundation models have been developed. RemoteCLIP improves image–text alignment for remote sensing by constructing large-scale remote sensing image–text pairs and demonstrates stronger zero-shot recognition and retrieval ability in this domain [44]. GeoRSCLIP and SkySense further explore remote sensing image–text alignment with large-scale geospatial data [45], [46]. SegEarth-OV further adapts open-vocabulary semantic segmentation to remote sensing images in a training-free manner, producing pixel-level semantic responses for arbitrary textual categories [19]. These studies show that vision–language priors can provide flexible semantic grounding for remote sensing interpretation [47], [48], [49], [50].

Despite their strong open-vocabulary recognition ability, most existing vision–language models are designed for single-temporal perception. They can identify or segment land-cover categories in an individual image, but they do not explicitly model how semantic states evolve across temporal observations. For change detection, the key problem is not only to recognize semantic categories at each temporal phase, but also to determine whether and how these categories transform over time. Therefore, vision–language models provide useful semantic priors for OVCD, but additional temporal reasoning is required to convert single-temporal open-vocabulary perception into reliable bi-temporal semantic change understanding.

C. Open-Vocabulary Change Detection

OVCD has recently emerged to overcome the fixed-label limitation of conventional CD methods by detecting changes associated with arbitrary textual or semantic concepts. Early foundation-model-based studies explore zero-shot or training-free change detection by combining segmentation, comparison, and semantic recognition modules. For example, AnyChange exploits SAM-based latent matching for class-agnostic change localization [51], while SCM uses semantic-aware foundation-model responses for unsupervised category-specific change detection [52]. DynamicEarth formally introduces OVCD and proposes two representative training-free paradigms, namely mask proposal–comparator–identifier (M–C–I) and identifier–mask proposal–comparator (I–M–C), which reuse off-the-shelf foundation models for proposal generation, temporal comparison, and semantic identification [20]. Following this modular direction, recent methods further explore different ways of integrating segmentation models, visual representation models, and vision–language models for open-vocabulary change localization.

Several recent studies have improved OVCD by enhancing change localization, model collaboration, and semantic reliability. Seg2Change adapts open-vocabulary semantic segmentation models to change detection by learning category-agnostic change maps and indexing bi-temporal semantic predictions with the detected changes [21]. AdaptOVCD develops a training-free architecture based on adaptive information fusion across data, feature, and decision levels, aiming to mitigate error propagation among heterogeneous pretrained models [22]. CoRegOVCD reformulates concept-specific change as calibrated posterior discrepancy and improves semantic reliability and spatial coherence through posterior calibration, structural verification, and regional consensus [23]. Recent works also investigate referring or prompt-driven change detection, where user-specified textual descriptions are used to guide change localization [24], [53]. These studies advance OVCD from simple model composition toward more reliable open-vocabulary change inference.

Despite their effectiveness, existing OVCD methods mainly infer change semantics from masks, dense concept responses,

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or separately predicted temporal endpoints. In such formulations, semantic changes are usually obtained by comparing candidate regions, measuring cross-temporal response discrepancies, or composing the semantic predictions of two temporal phases. Although these designs enable flexible reuse of foundation models, they do not explicitly represent a semantic change as a directed from-to transition. Consequently, endpoint errors or intermediate mask errors may propagate to the final change result, and the temporal direction of land-cover evolution is only indirectly modeled. Different from these mask-, response-, or endpoint-centric formulations, this work studies OVCD from a transition-centric perspective. We explicitly represent semantic changes as directed $i \rightarrow j$ transitions and directly predict pixel-level transition responses from bi-temporal observations, enabling unified reasoning over changed regions and their from-to semantics.

III. Method

A. Problem Formulation

Given a pair of bi-temporal remote sensing images $I_1, I_2 \in \mathbb{R}^{H\times W\times 3}$ acquired over the same geographic area and an open-vocabulary prompt set $\mathcal{V} = \{v_1, v_2, \ldots, v_N\}$, OVCD aims to identify land-cover changes associated with semantic concepts specified by the textual vocabulary $\mathcal{V}$. Existing OVCD methods usually infer change semantics by comparing masks, measuring temporal response discrepancies, or composing semantic predictions from the two temporal phases. In contrast, we formulate OVCD as an explicit semantic transition prediction problem, where each changed pixel is assigned a directed transition $v_i \rightarrow v_j$ from the prechange semantic state to the postchange semantic state. All unchanged pixels are assigned to a unified no-change state. Accordingly, the transition vocabulary is defined as

$$\mathcal{T} = \{0\} \cup \{(i, j) \mid i, j \in \{1, \ldots, N\}, i \neq j\} \tag{1}$$

where 0 denotes no-change and each nonzero element represents an explicit directed semantic transition.

This transition-centric formulation differs from conventional endpoint composition mainly in the final decision mechanism. Endpoint-based methods first estimate the semantic states of the two temporal phases and then compose them into a transition label. Such an indirect process may accumulate endpoint errors, since an incorrect prediction at either temporal phase can lead to an incorrect transition. Moreover, land-cover evolution is inherently directional, where $v_i \rightarrow v_j$ and $v_j \rightarrow v_i$ usually correspond to different geographic processes. In our framework, endpoint semantic distributions are still used to construct a compact candidate transition set, but they do not directly determine the final transition label. Instead, each candidate transition is explicitly scored using joint bi-temporal evidence. Therefore, endpoint semantics serve as a candidate prior, while the final decision is made at the transition level rather than through hard endpoint composition.

B. Framework Overview

The overall framework is illustrated in Fig. 2. It consists of three main stages: open-vocabulary representation extraction, proposal-guided explicit transition prediction, and consistency-aware semantic transition refinement. First, the frozen DINOv2 and SegEarth-OV3 are used to provide complementary visual and semantic priors. DINOv2 extracts bi-temporal visual features for change proposal generation and transition encoding, while SegEarth-OV3 produces two distinct outputs: semantic probability maps and text prototypes. The semantic probability maps provide pixel-level category probabilities, whereas the text prototypes provide language-aligned category embeddings for constructing transition prototypes. These foundation models remain frozen, and their outputs are precomputed and cached before transition training.

Second, a lightweight proposal branch predicts a dense change probability map from the bi-temporal DINOv2 features. The resulting proposals identify ROIs that may contain changes and are used only as region indices rather than as final change masks. For each ROI, DINOv2 visual features and SegEarth-OV3 semantic probability maps are cropped and aligned to construct an ROI-level query representation. Based on the ROI semantic distributions of the two temporal phases, a compact Top-k candidate transition set C is initialized by combining the most likely from-to endpoints. For example, the candidate set of an ROI may contain "vegetation to building," "bareland to building," and no-change. Explicit transition prototypes are then constructed from SegEarth-OV3 text prototypes and selected according to C. By matching the ROI query map with the selected transition prototypes, the model produces pixel-level transition logits, which denote the unnormalized scores of the candidate transitions before softmax normalization.

Third, two components are introduced to improve candidate coverage and transition reliability. The RCSM predicts a residual Top-m transition set and supplements it into C, reducing candidate omission caused by uncertain endpoint semantic distributions. The CSTRM further refines transition logits through TMC and TSR. TMC calibrates each candidate transition according to its support from the corresponding prechange and postchange semantic marginals. For example, a high "vegetation to building" score should be supported by strong prechange evidence for vegetation and strong postchange evidence for building. TSR improves spatial coherence by reducing isolated and fragmented predictions within the same changed region. Finally, ROI-level sparse transition logits are stitched back to the image space to obtain the global explicit transition map.

C. Foundation Representation and Proposal ROI Encoding

As shown in the left part of Fig. 2, we first use frozen foundation models to provide visual and semantic representations for explicit transition prediction. Given the bi-temporal images I_1 and I_2, DINOv2 extracts multilevel visual features

$$\mathcal{F}_1^{\mathrm{D}} = \left\{F_1^{\mathrm{D},l}\right\}_{l\in\mathcal{L}}, \quad \mathcal{F}_2^{\mathrm{D}} = \left\{F_2^{\mathrm{D},l}\right\}_{l\in\mathcal{L}} \tag{2}$$

where $\mathcal{L} = \{2, 5, 8, 11\}$ denotes the selected encoder layers. Meanwhile, SegEarth-OV3 takes the prompt

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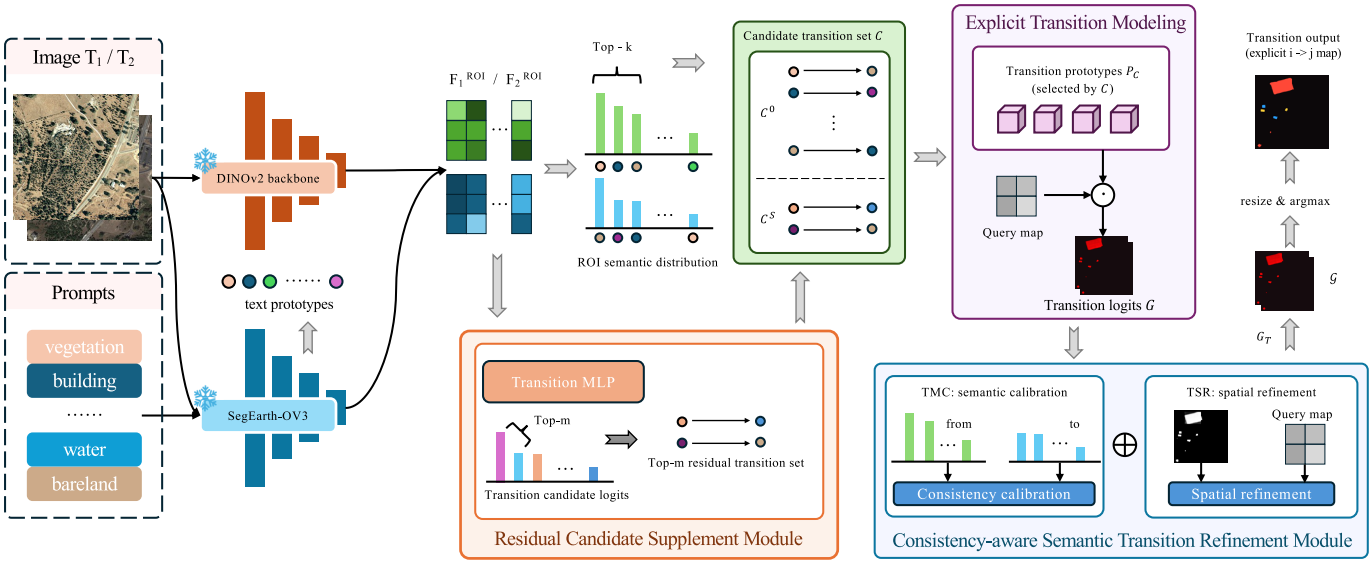


Fig. 2. Overview of the proposed framework. Frozen DINOv2 and SegEarth-OV3 provide bi-temporal visual-semantic representations for each candidate ROI. Top-k endpoint combinations initialize a compact candidate transition set, which is supplemented by RCSM. Explicit transition modeling predicts sparse $i \rightarrow j$ logits, and CSTRM performs semantic calibration and spatial refinement before reconstruction of the global transition map.

vocabulary $\mathcal{V} = \{v_1, \ldots, v_N\}$ as input and produces open-vocabulary semantic probability maps

$$\mathcal{F}_1^{\mathrm{S}}, \mathcal{F}_2^{\mathrm{S}} \in \mathbb{R}^{N \times H \times W} \tag{3}$$

as well as text prototypes

$$\mathbf{E} = \{\mathbf{e}_i\}_{i=1}^{N}, \quad \mathbf{e}_i \in \mathbb{R}^{d_s}. \tag{4}$$

The semantic probability maps describe the probability of each prompted category at every spatial location, whereas each text prototype e_i is the language-aligned embedding of category v_i. Both DINOv2 and SegEarth-OV3 are kept frozen, and their outputs are precomputed and cached before transition training. DINOv2 features provide bi-temporal feature differences, while SegEarth-OV3 semantic maps and text prototypes provide open-vocabulary endpoint priors and semantic anchors for transition prototype construction.

The explicit transition formulation enlarges the prediction space from semantic categories to directed semantic transitions. For a prompt vocabulary with N categories, the transition space contains one no-change state and $N(N-1)$ changed transition states. Taking the SECOND dataset as an example, its six land-cover categories lead to $1+6\times5 = 31$ transition states. This quadratic growth is a computational cost introduced by explicit transition modeling. Therefore, we restrict transition prediction to candidate ROIs and compact candidate transition sets instead of densely scoring the complete transition vocabulary at every pixel.

Given the cached DINOv2 features $\mathcal{F}_1^{\mathrm{D}}$ and $\mathcal{F}_2^{\mathrm{D}}$, the proposal branch first computes layer-wise temporal difference features

$$D^l = \left| F_2^{\mathrm{D},l} - F_1^{\mathrm{D},l} \right|, \quad l \in \mathcal{L}. \tag{5}$$

The difference features from different layers are projected into a shared feature space, spatially aligned, and fused by a lightweight proposal head to obtain a dense change probability map $M^p \in [0,1]^{H \times W}$. The probability map is further processed by thresholding, morphological filtering, connected component extraction, and region ranking to obtain a set of candidate ROI regions

$$\mathcal{R} = \{r_q\}_{q=1}^{Q}. \tag{6}$$

Here, each r_q denotes the bounding box of the qth ROI, together with its associated proposal mask. These proposals are not used as final change masks, but rather serve as region indices that guide subsequent transition prediction.

For clarity, we describe the following operations on a generic ROI $r \in \mathcal{R}$ and omit the ROI index unless otherwise specified. DINOv2 features from the two temporal phases are cropped according to r and resized to a fixed spatial size $h \times w$, yielding ROI visual feature sets

$$\mathcal{U}_1^{\mathrm{D}} = \left\{U_1^{\mathrm{D},l}\right\}_{l\in\mathcal{L}}, \quad \mathcal{U}_2^{\mathrm{D}} = \left\{U_2^{\mathrm{D},l}\right\}_{l\in\mathcal{L}} \tag{7}$$

where $U_1^{\mathrm{D},l}$ and $U_2^{\mathrm{D},l}$ are the ROI features cropped from the lth DINOv2 layer. Similarly, the SegEarth-OV3 semantic maps are cropped from the same image-space region and resized to obtain ROI semantic maps

$$U_1^{\mathrm{S}}, U_2^{\mathrm{S}} \in \mathbb{R}^{N \times h \times w} \tag{8}$$

where U_1^{S} and U_2^{S} denote the ROI-level semantic probability maps of the two temporal phases. These semantic maps correspond to the ROI semantic distributions in Fig. 2 and provide local endpoint priors for candidate transition construction.

The ROI visual feature sets $\mathcal{U}_1^{\mathrm{D}}$ and $\mathcal{U}_2^{\mathrm{D}}$ are further projected and fused into compact visual representations Z_1 and Z_2. To encode transition-aware bi-temporal evidence, we construct a query representation from the two temporal features, their difference, their absolute difference, their element-wise interaction, and the ROI semantic maps. The resulting ROI query map is denoted as

$$Q \in \mathbb{R}^{d \times h \times w}. \tag{9}$$

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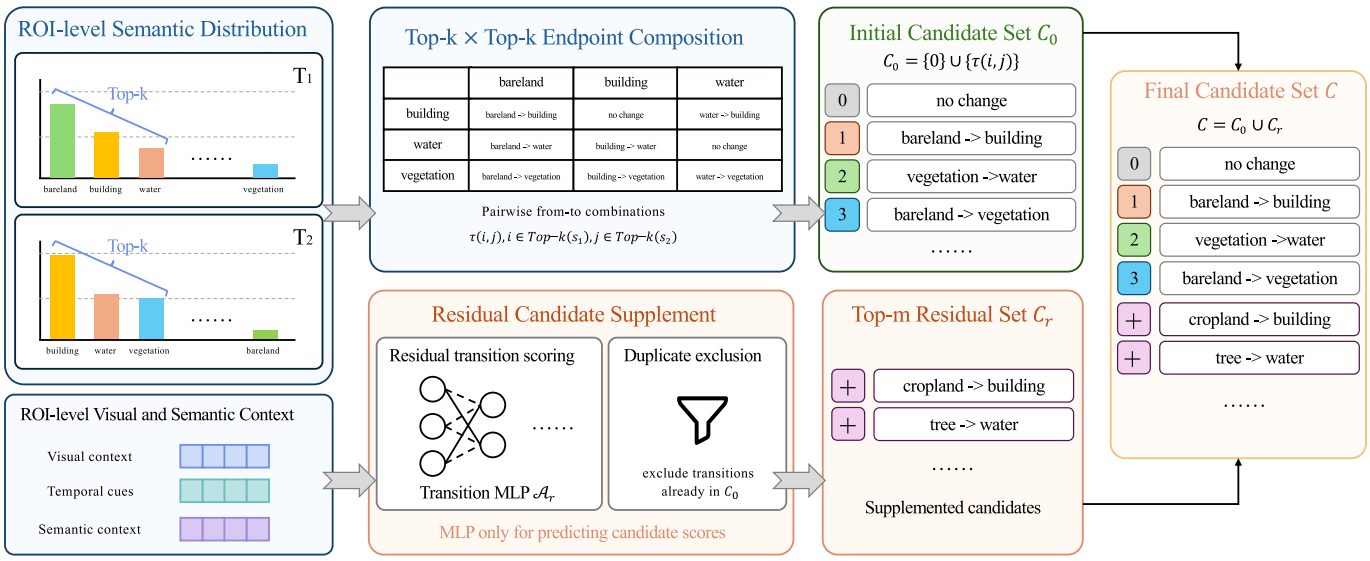


Fig. 3. Illustration of the two-branch candidate construction in RCSM. The upper branch forms the initial candidate set C^0 from Top-k endpoint combinations. The lower branch scores transitions outside C^0 and selects the Top-m residual set C^r. The two sets are merged to form the final candidate set $C = C^0 \cup C^r$.

Here, Q corresponds to the query map in Fig. 2. It integrates bi-temporal feature differences, temporal interactions, and open-vocabulary semantic priors, and is used for explicit transition scoring and transition-aware refinement.

D. Candidate Transition Set Construction With RCSM

Before explicit transition scoring, we first construct a compact candidate transition set for each ROI. This step reduces the large open-vocabulary transition space to a small set of locally plausible transitions while preserving the transition states that are likely to occur within the ROI. The initial candidate set is derived from ROI-level semantic endpoint distributions.

Given the ROI semantic maps U_1^S and U_2^S, we spatially average them to obtain endpoint-level semantic responses

$$\mathbf{s}_1 = \mathrm{AvgPool}\left(U_1^S\right), \quad \mathbf{s}_2 = \mathrm{AvgPool}\left(U_2^S\right). \tag{10}$$

Here, $\mathbf{s}_1, \mathbf{s}_2 \in \mathbb{R}^N$ denote the ROI-level semantic distributions of the two temporal phases, where each dimension corresponds to a prompt category in $\mathcal{V}$. We then select the Top-k categories from each distribution and compose their pairwise directed transitions

$$C^0 = \{0\} \cup \left\{\tau(i,j) \mid i \in \mathrm{TopK}(\mathbf{s}_1),\ j \in \mathrm{TopK}(\mathbf{s}_2),\ i \neq j\right\}. \tag{11}$$

In this formulation, 0 denotes the no-change state, and $\tau(i,j)$ maps the directed semantic pair $v_i \rightarrow v_j$ to its global transition index in $\mathcal{T}$. The resulting set C^0 corresponds to the Top-k candidate transition set shown in Fig. 2. It retains locally plausible from-to combinations while avoiding dense scoring over the full transition vocabulary.

However, C^0 is initialized from marginal endpoint semantic responses. When either temporal endpoint is uncertain or ranked outside the Top-k categories, the corresponding true transition may be excluded before explicit transition scoring. This candidate omission problem becomes more pronounced in open-vocabulary scenarios, where a large prompt vocabulary introduces more semantically related categories and increases the difficulty of reliable endpoint selection.

To alleviate this issue, we introduce RCSM, as detailed in Fig. 3. Instead of replacing the initial candidate set, RCSM predicts additional residual transition candidates that are not already included in C^0. Specifically, RCSM estimates omitted changed transitions from pooled ROI-level visual-semantic context. Since the no-change state is always retained in C^0, RCSM focuses only on changed transition states. The residual candidate set is obtained by selecting the Top-m transitions according to the residual scores

$$C^r = \mathrm{TopIdx}_m\left(\left\{\mathcal{A}_{r,t}\right\}_{t \in \mathcal{T}^+,\ t \notin C^0}\right) \tag{12}$$

where $\mathcal{A}_{r,t}$ denotes the residual score of changed transition t, $\mathrm{TopIdx}_m(\cdot)$ returns the transition indices corresponding to the m largest residual scores, $\mathcal{T}^+$ is the changed transition space, and m is the number of supplemented transitions. By excluding transitions already included in C^0, RCSM only contributes residual candidates. The final candidate transition set is then formed as

$$C = C^0 \cup C^r. \tag{13}$$

In this way, RCSM supplements potentially missing transitions while preserving the reliable endpoint-based candidates in C^0. The resulting candidate set C is used to select transition prototypes and perform explicit pixel-level transition scoring in the next stage.

E. Explicit Semantic Transition Modeling

Given the final candidate transition set C, explicit semantic transition modeling predicts pixel-level scores for the candidate transitions within each ROI. Here, the candidate transition set refers to a compact collection of locally plausible directed transitions, such as "vegetation to building," "bareland to building," and no-change. Although C is partly constructed

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from endpoint semantic distributions, the model does not select two endpoint labels and directly compose them into the final transition. Instead, each candidate transition $v_i \rightarrow v_j$ is explicitly scored using its language-aligned transition prototype and the joint bi-temporal query map Q. Accordingly, the final output is determined by transition-level scores within the candidate set rather than by hard endpoint composition. Before softmax normalization, these transition-level scores are referred to as transition logits, whereas the normalized values are referred to as transition probabilities.

We first construct a global transition prototype bank from the SegEarth-OV3 text prototypes $\mathbf{E} = \{\mathbf{e}_i\}_{i=1}^{N}$. For each changed transition $v_i \rightarrow v_j$, where $i \neq j$, its transition prototype is generated as

$$\mathbf{p}_{i\rightarrow j} = g_\theta\left(\left[\mathbf{e}_i, \mathbf{e}_j, \mathbf{e}_j - \mathbf{e}_i, \mathbf{e}_i \odot \mathbf{e}_j\right]\right) \tag{14}$$

where g_θ denotes the transition prototype MLP, $[\cdot]$ is channel-wise concatenation, and $\odot$ denotes element-wise multiplication. The four terms encode the source semantics, target semantics, directional semantic shift, and endpoint interaction. The generated prototypes are normalized and organized into a global transition prototype bank

$$\mathbf{P} = \{\mathbf{p}_t\}_{t=0}^{|\mathcal{T}|-1}, \quad \mathbf{p}_t \in \mathbb{R}^d \tag{15}$$

where $\mathbf{p}_0$ is implemented as a learnable no-change prototype and $\mathbf{p}_t$ for $t > 0$ represents a changed transition prototype.

For the current ROI, only the prototypes indexed by the final candidate set C are selected

$$\mathbf{P}_C = \{\mathbf{p}_t \mid t \in C\}. \tag{16}$$

The selected transition prototypes are then matched with the ROI query map Q through pixel-wise dot products

$$G_t = \mathbf{p}_t^{\top} Q, \quad t \in C \tag{17}$$

where $G_t \in \mathbb{R}^{h\times w}$ denotes the logit map of transition t. By stacking all candidate logit maps, we obtain sparse transition logits

$$G = \{G_t\}_{t\in C} \in \mathbb{R}^{|C|\times h\times w}. \tag{18}$$

Since C is a sparse subset of the full transition vocabulary, each local channel of G corresponds to a global transition index in C. In this way, the model directly produces explicit $i \rightarrow j$ transition responses from bi-temporal ROI evidence, without using endpoint composition as the final prediction rule.

F. Consistency-Aware Semantic Transition Refinement

Although direct transition prediction produces transition-specific logits, the predicted responses may still suffer from semantic ambiguity and local spatial inconsistency. A reliable transition prediction should satisfy two requirements. First, each transition response should be consistent with the source and target marginals induced by the candidate transition distribution. For example, a high prediction score for "vegetation to building" should be supported by a high aggregated probability for vegetation as the prechange category and building as the postchange category. Second, the transition logits should be spatially coherent within changed regions, meaning that pixels belonging to the same changed region should not produce isolated or fragmented predictions. To this end, we introduce CSTRM, which consists of TMC and TSR. As shown in Fig. 4, CSTRM refines the initial transition logits through TMC and TSR.

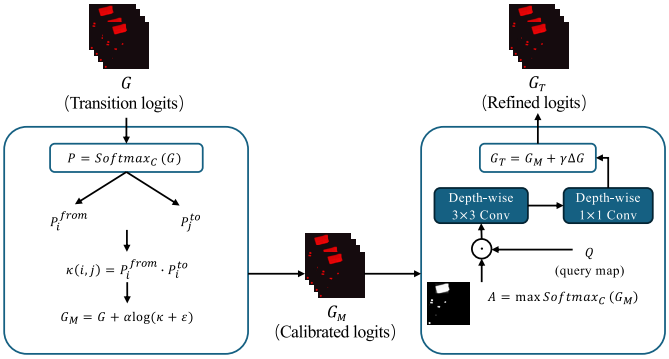


Fig. 4. Illustration of CSTRM. TMC calibrates transition logits by TMC, and TSR refines them through confidence-guided query refinement and candidate-conditioned projection.

1) Transition-Marginal Consistency: Given the sparse transition logits $G \in \mathbb{R}^{|C|\times h\times w}$, TMC first normalizes them over the candidate transition set

$$P = \mathrm{Softmax}_C(G) \tag{19}$$

where $P \in \mathbb{R}^{|C|\times h\times w}$ denotes the transition probability maps over C. Since each changed transition corresponds to a directed endpoint pair, we aggregate transition probabilities over changed candidates into from- and to-endpoint marginal responses

$$P_i^{\mathrm{from}} = \sum_{t\in C,\ t\neq 0,\ \mathrm{from}(t)=i} P_t, \quad P_j^{\mathrm{to}} = \sum_{t\in C,\ t\neq 0,\ \mathrm{to}(t)=j} P_t \tag{20}$$

where from(t) and to(t) denote the source and target endpoint indices of transition t. For a changed transition $t = \tau(i, j)$, its transition-marginal compatibility is computed as

$$\kappa_t = P_i^{\mathrm{from}} \cdot P_j^{\mathrm{to}}. \tag{21}$$

The compatibility scores of all changed candidate transitions form a compatibility map κ. For changed transitions, TMC calibrates the logits as

$$G_{M,t} = G_t + \alpha \log(\kappa_t + \epsilon), \quad t \in C,\ t \neq 0 \tag{22}$$

where $G_{M,t}$ denotes the TMC-calibrated logit map of transition t, α is a learnable scaling parameter controlling the calibration strength, and ϵ is a small constant for numerical stability. The no-change logit is kept unchanged, and the calibrated logits are collectively denoted by G_M. Through TMC, transition responses are calibrated according to the support provided by their induced source and target marginals. It should be noted that TMC performs candidate-wise semantic calibration without spatial convolution or cross-pixel interaction, whereas spatial refinement is introduced separately by TSR.

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2) Transition-Aware Spatial Refinement: TMC improves TMC, but pixel-level transition logits may still be spatially fragmented. To enhance regional coherence, TSR refines the TMC-calibrated logits by using transition-aware confidence to guide spatial correction. Specifically, we derive a confidence map from G_M

$$A(x) = \max_{t \in C} \mathrm{Softmax}_C(G_M)_t(x) \tag{23}$$

where $A \in [0,1]^{h \times w}$ reflects the confidence of the most likely transition at each ROI location. The confidence map highlights reliable transition regions and suppresses uncertain responses.

TSR first uses A to produce a spatially refined query residual

$$\Delta Q = \phi_{\mathrm{tsr}}(A \odot Q) \tag{24}$$

where A is broadcast along the channel dimension of Q and ϕ_{tsr} is a lightweight depthwise-pointwise convolutional branch. The logit residual is then computed by candidate-conditioned projection

$$\Delta G_t(x) = \langle \psi(\Delta Q(x)), \mathbf{p}_t \rangle, \quad t \in C \tag{25}$$

where $\psi(\cdot)$ maps ΔQ to the transition prototype space. The final TSR-refined logits are

$$G_T(t,x) = G_M(t,x) + \gamma \Delta G_t(x), \quad t \in C \tag{26}$$

where γ is a learnable scalar controlling the refinement strength. Since ΔG_t is computed for $t \in C$, its dimension naturally matches the sparse candidate set $|C|$. TSR does not change C; it only refines logits within the selected candidate space.

G. Prediction Structure and Loss Function

1) Prediction Structure: After CSTRM, the refined ROI-level transition logits G_T are projected back to the image space to generate the final prediction. Since each ROI is scored only over its candidate transition set C, the local candidate channels are first mapped to their corresponding global transition indices in $\mathcal{T}$. We maintain a global transition score volume

$$\mathcal{S} \in \mathbb{R}^{|\mathcal{T}| \times H \times W} \tag{27}$$

where each channel corresponds to one transition state in the global transition vocabulary. The no-change channel is initialized as a neutral background score, while changed transition channels are initialized with low scores before ROI write-back.

For each ROI $r \in \mathcal{R}$, the refined logits G_T are resized from the ROI resolution $h \times w$ to the spatial size of the corresponding ROI region. Then, each local candidate logit map is written back to the global transition score volume according to its global transition index. For overlapping ROIs, we use max-score fusion for each transition channel

$$\mathcal{S}_t(x) = \max\left(\mathcal{S}_t(x), \mathrm{Resize}\left(G_{T,t}\right)(x)\right), \quad t \in C. \tag{28}$$

Pixels not covered by any ROI remain assigned to the no-change state. The final explicit transition map is obtained by selecting the transition state with the highest global score at each pixel

$$\hat{Y}^{\mathrm{tr}}(x) = \arg\max_{t \in \mathcal{T}} \mathcal{S}_t(x). \tag{29}$$

Here, $\hat{Y}^{\mathrm{tr}}(x) = 0$ denotes no-change, while $\hat{Y}^{\mathrm{tr}}(x) = \tau(i,j)$ represents $v_i \to v_j$. Projection yields binary masks and class-wise semantic change maps. For changed pixels, source and target endpoint labels are also read from the transition state; unchanged regions remain a unified no-change state and do not recover full endpoint maps.

2) Loss Function: The training objective consists of a proposal loss and a transition prediction loss. The proposal branch is supervised by the binary change mask

$$\mathcal{L}_{\mathrm{prop}} = \mathrm{BCE}\left(M^p, Y^{\mathrm{bin}}\right) \tag{30}$$

where $Y^{\mathrm{bin}} \in \{0,1\}^{H \times W}$ denotes the binary change label.

For datasets with explicit semantic transition annotations, Y^{tr} is constructed from the two temporal semantic labels. The final candidate set is formed as $C = C^0 \cup C^r$, where C^0 is obtained by Top-k endpoint composition and C^r is supplemented by RCSM. Since the prediction head only scores transitions in C, the sparse logits $G_T^C \in \mathbb{R}^{|C| \times h \times w}$ are scattered back to the global transition space for loss computation

$$\bar{G}_{T,t}(x) = G_{T,\pi_C(t)}(x), \quad t \in C \tag{31}$$

where $\pi_C(t)$ denotes the local index of transition t in C. Transitions not included in C are masked out. We define a candidate-coverage mask

$$\mathcal{M}_C(x) = \mathbb{I}\left[Y^{\mathrm{tr}}(x) \in C\right]. \tag{32}$$

The explicit transition loss is computed only on covered pixels

$$\mathcal{L}_{\mathrm{tr}} = \frac{1}{\sum_x \mathcal{M}_C(x)} \sum_x \mathcal{M}_C(x)\, \mathrm{CE}\left(\bar{G}_T(x), Y^{\mathrm{tr}}(x)\right). \tag{33}$$

When explicit transition supervision is available, RCSM is supervised to recover changed transitions omitted by C^0. Let $\mathcal{Y}^{\mathrm{tr}}$ denote the ROI-level ground-truth changed transition set. The residual target is

$$y_t^r = \begin{cases} 1, & t \in \mathcal{Y}^{\mathrm{tr}},\ t \notin C^0,\ t \neq 0 \\ 0, & \text{otherwise} \end{cases} \qquad t \in \mathcal{T}^+. \tag{34}$$

The auxiliary RCSM loss is

$$\mathcal{L}_{\mathrm{rcsm}} = \mathrm{BCE}\left(\mathcal{A}_r, y^r\right) \tag{35}$$

where $\mathcal{A}_r$ denotes the residual transition scores predicted by RCSM.

For binary-supervised datasets, explicit transition labels are unavailable and $\mathcal{L}_{\mathrm{rcsm}}$ is omitted. The residual scores of selected transitions are added to their logits

$$\widetilde{G}_{T,t} = G_{T,t} + \mathbb{I}\left[t \in C^r\right] \mathcal{A}_{r,t}, \quad t \in C. \tag{36}$$

Binary logits are obtained by aggregating changed candidates

$$G_T^{\mathrm{bin}} = \left[\widetilde{G}_{T,0},\ \mathrm{LogSumExp}_{t \in C,\, t \neq 0}\left(\widetilde{G}_{T,t}\right)\right] \tag{37}$$

and the transition loss is

$$\mathcal{L}_{\mathrm{tr}} = \mathrm{CE}\left(G_T^{\mathrm{bin}}, Y^{\mathrm{bin}}\right). \tag{38}$$

Thus, RCSM is indirectly optimized through the selected residual scores under binary supervision.

The final objective is

$$\mathcal{L} = \lambda_{\mathrm{prop}} \mathcal{L}_{\mathrm{prop}} + \lambda_{\mathrm{tr}} \mathcal{L}_{\mathrm{tr}} + \lambda_{\mathrm{rcsm}} \mathcal{L}_{\mathrm{rcsm}} \tag{39}$$

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where $\mathcal{L}_{\text{rcsm}}$ is used only with explicit transition supervision. For binary-only datasets, $\lambda_{\text{rcsm}} = 0$. During inference, the candidate set is constructed without using ground-truth information.

IV. Experiments

A. Datasets and Benchmarks

We evaluate our method on six widely used remote sensing change detection benchmarks, following the settings of recent OVCD studies. WHU-CD [32] and LEVIR-CD [3] are used for building change detection, DSIFN [34] and CLCD [54] for land-cover change detection, and SC-SCD7 [55] and SECOND [14] for SCD. These benchmarks cover different scenarios from single-category object changes to multicategory land-cover changes and semantic transitions.

The dataset splits used in our experiments are as follows. WHU-CD contains 1260 train and 690 test image pairs of size 512×512. LEVIR-CD contains 445/64/128 source image pairs for train/validation/test, respectively, with an original resolution of 1024×1024; following the adopted preprocessing protocol, each pair is divided into four nonoverlapping 512×512 crops, resulting in 1780/256/512 pairs used by the data loader. DSIFN contains 3600/340/48 train/validation/test pairs, and CLCD contains 360/120/120 pairs, all at 512×512 resolution. For SC-SCD7, the original 1400 train pairs are deterministically divided into 1260 train and 140 validation pairs, with an additional 322 test pairs. SECOND contains 2374 unique train pairs and 594 test pairs, covering all 2968 image pairs. WHU-CD, LEVIR-CD, DSIFN, and CLCD provide binary change annotations; therefore, explicit transition labels are unavailable for these datasets. SC-SCD7 and SECOND provide bi-temporal semantic labels, from which explicit transition labels are directly constructed. Unchanged pixels are assigned to the no-change state, while changed pixels are mapped to directed semantic transitions.

Following the benchmark-level evaluation practice of recent OVCD studies [20], [21], quantitative evaluation is conducted within the annotated semantic space available in each dataset. Therefore, the reported results evaluate the proposed prompt-conditioned transition formulation on the corresponding benchmark categories rather than under a strict category-disjoint seen/unseen protocol.

For fair comparison with existing OVCD methods, we report changed-class metrics on binary benchmarks, including F1^{c}, IoU^{c}, and Kappa. For SCD benchmarks, we report SCD metrics, including mF1^{c}, mIoU^{c}, mOA, and mKappa.

B. Experimental Setup

We conduct all experiments using the PyTorch framework on NVIDIA GeForce RTX 3090 GPUs with 24-GB memory. Each training run is performed on a single GPU. All transition models are trained for 50 epochs using the AdamW optimizer with an initial learning rate of 0.0002 and a weight decay of 0.0001. We use a batch size of 4 for SECOND and SC-SCD7, and a batch size of 16 for LEVIR-CD, WHU-CD, DSIFN, and CLCD. Frozen DINOv2 [25] and SegEarth-OV3 [26] features are cached for transition training. DINOv2 features from layers $\{2, 5, 8, 11\}$ are cached as multilevel visual features, while SegEarth-OV3 provides prompt-conditioned semantic probability maps and text prototypes. Both foundation models are kept frozen during training.

TABLE I

Prompt Class Names Used for the Evaluation Datasets. Braces Indicate Multiple Prompt Words for a Single Category. Foreground Evaluation Classes Are Highlighted in Bold

Dataset	Category	Class Name
WHU-CD [32]	Building	background, {bareland, barren}, grass, car, {tree, forest}, {water, river}, cropland, **{building, roof, house}**
LEVIR-CD [3]	Building	background, {bareland, barren}, grass, car, {tree, forest}, {water, river}, cropland, **{building, roof, house}**
DSIFN [34]	Land-Cover	background, **{building, roof, house}**, **garden**, **playground**, **{construction, apartment, residential}**, **materials**, **{tree, forest}**, **{water, river}**
CLCD [54]	Land-Cover	background, **{bareland, barren, grass}**, **car**, **{tree, forest}**, **{water, river}**, **cropland**, **{building, roof, house}**
SC-SCD7 [55]	Bareland	**bareland, barren, ground, floor, soil**
	Water	**water, river, pond**
	Building	**building, roof, house**
	Structure	**structure, construction, greenhouse**
	Farmland	**farmland, cropland, terrace**
	Vegetation	**grass, vegetation, tree, forest**
	Road	**road**
SECOND [14]	Water	**water, river**
	Ground	**bareland, barren, ground**
	Low vegetation	**grass, cropland**
	Tree	**tree, forest**
	Building	**building, roof, house**
	Playground	**sports field**

The prompt vocabularies used for SegEarth-OV3 are listed in Table I. Multiple prompt words belonging to the same semantic category are grouped as synonyms and mapped to a single endpoint category.

The initial candidate transition set is constructed by selecting the Top-k semantic categories from each temporal endpoint distribution and forming their Cartesian product, with the no-change state always retained.

In the full model, RCSM supplements the initial candidate set and CSTRM refines candidate transition logits. The (k, m) settings are $(4, 6)$ for WHU-CD, $(5, 6)$ for LEVIR-CD, $(4, 4)$ for DSIFN, $(6, 4)$ for CLCD, $(4, 4)$ for SC-SCD7, and $(5, 8)$ for SECOND. The Top-k sensitivity study disables RCSM/CSTRM. The model is trained with AdamW using a learning rate of 2×10^{-4}, a weight decay of 1×10^{-4}, and a fixed random seed of 3407. We set $\lambda_{\text{prop}} = 1.0$, $\lambda_{\text{tr}} = 1.0$, and $\lambda_{\text{rcsm}} = 0.3$ for explicit transition supervision or 0 for binary supervision.

During inference, the candidate set is constructed only from Top-k endpoint composition and RCSM supplementation, without using any ground-truth information. Predictions are projected to the required evaluation space: binary masks for

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TABLE II

QUANTITATIVE RESULTS FOR STATE-OF-THE-ART COMPARISON ON BINARY AND LAND-COVER CHANGE DETECTION BENCHMARKS. PERCENTAGE METRICS (%) ARE REPORTED FOR $F1^c$ AND IoU^c, WHILE KAPPA IS REPORTED AS A COEFFICIENT. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY

Method	WHU-CD			LEVIR-CD			DSIFN			CLCD		
	$F1^c$	**IoU^c**	**Kappa**	**$F1^c$**	**IoU^c**	**Kappa**	**$F1^c$**	**IoU^c**	**Kappa**	**$F1^c$**	**IoU^c**	**Kappa**
Traditional unsupervised CD												
PCA-KMeans [28]	14.33	7.72	0.0846	9.96	5.24	0.0116	35.48	21.57	0.1865	18.42	10.14	0.0780
CVA [1]	7.17	3.72	0.0023	9.27	4.86	0.0017	27.88	16.20	0.0142	13.39	7.17	0.0017
DCVA [31]	20.55	11.45	0.1537	13.85	7.44	0.0507	38.52	23.85	0.1758	20.81	11.62	0.0918
Foundation-model-based training-free CD												
SCM [52]	32.13	19.14	0.2792	32.36	19.30	0.2734	40.13	25.10	0.2443	23.31	13.19	0.1522
AnyChange [51]	28.13	16.37	0.2330	32.68	19.53	0.2672	39.19	24.37	0.2280	31.96	19.02	0.2342
Inst-CEG [56]	62.54	45.49	0.6074	63.29	46.30	0.6084	31.81	18.91	0.2430	6.76	3.50	0.0020
OVCD												
DynamicEarth (M-C-I) [20]	57.35	40.20	0.5541	46.43	30.23	0.4242	54.35	37.32	0.4741	23.83	13.52	0.1916
DynamicEarth (I-M-C) [20]	75.85	61.09	0.7488	69.70	53.50	0.6789	26.42	15.22	0.2260	14.97	8.09	0.1382
Seg2Change [21]	85.96	75.38	0.8541	78.71	64.89	0.7741	58.61	41.46	0.5085	47.88	31.48	0.4238
Ours	**91.64**	**84.56**	**0.9133**	**83.32**	**71.42**	**0.8257**	**72.59**	**56.97**	**0.5816**	**77.14**	**62.79**	**0.7526**

TABLE III

QUANTITATIVE RESULTS FOR STATE-OF-THE-ART COMPARISON ON SCD BENCHMARKS. PERCENTAGE METRICS (%) ARE REPORTED FOR $mF1^c$, $mIoU^c$, AND mOA, WHILE mKAPPA IS REPORTED AS A COEFFICIENT. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY

Method	SC-SCD7				SECOND			
	$mF1^c$	**$mIoU^c$**	**mOA**	**mKappa**	**$mF1^c$**	**$mIoU^c$**	**mOA**	**mKappa**
Foundation-model-based training-free CD								
SCM [52]	12.06	6.51	85.95	0.0619	14.63	8.40	78.61	0.0795
AnyChange [51]	16.51	9.32	78.27	0.1056	19.54	11.84	76.07	0.1252
Inst-CEG [56]	6.90	3.68	94.40	0.0551	17.82	10.46	92.80	0.1604
OVCD								
DynamicEarth (M-C-I) [20]	29.11	17.97	91.37	0.2532	37.51	23.58	91.88	0.3297
DynamicEarth (I-M-C) [20]	9.19	5.16	95.25	0.0877	22.17	13.72	93.76	0.2109
Seg2Change [21]	35.70	23.23	95.82	0.3387	41.74	28.16	**95.08**	0.3928
Ours	**54.62**	**38.95**	**96.24**	**0.5260**	**50.50**	**34.57**	93.39	**0.4678**

WHU-CD, LEVIR-CD, DSIFN, and CLCD, and class-wise semantic change maps for SC-SCD7 and SECOND.

C. Comparison With State-of-the-Art Methods

We compare the proposed method with representative baselines from three groups: traditional unsupervised CD, foundation-model-based training-free CD, and OVCD. In particular, DynamicEarth [20] and Seg2Change [21] are included as representative existing OVCD methods and are evaluated using the same datasets and metrics. Open-vocabulary capability refers to recognizing prompt-defined categories at inference time and does not require the change detection component to be training-free. Therefore, existing OVCD methods include both training-free approaches, such as DynamicEarth, and trainable approaches, such as Seg2Change and our method. These methods are compared under their original protocols to reflect the current OVCD landscape, while Seg2Change provides the closest trainable baseline. The compared methods include PCA-KMeans [28], CVA [1], DCVA [31], SCM [52], AnyChange [51], Inst-CEG [56], DynamicEarth, and Seg2Change.

1) Quantitative Comparison on Benchmark Datasets: Table II reports the quantitative comparison on binary and land-cover change detection benchmarks. WHU-CD and LEVIR-CD mainly evaluate single-category building change detection, while DSIFN and CLCD involve more diverse land-cover changes. The proposed method achieves the best results on all four benchmarks. Specifically, it obtains 91.64%, 83.32%, 72.59%, and 77.14% $F1^c$ on WHU-CD, LEVIR-CD, DSIFN, and CLCD, respectively. Compared with the strongest competing method, Seg2Change, our method improves $F1^c$ by 5.68, 4.61, 13.98, and 29.26 percentage points on the four benchmarks. Similar improvements can also be observed in IoU^c and Kappa, indicating that the gains are not caused by a single metric bias. The improvements on DSIFN and CLCD are particularly evident, where multiple land-cover categories and complex background variations make endpoint- or mask-based comparison more ambiguous. These results show that directly modeling semantic transitions from bi-temporal observations provides more reliable change evidence than conventional unsupervised, training-free, or mask-centric OVCD pipelines.

Table III further compares different methods on SCD benchmarks. Compared with BCD, SC-SCD7 and SECOND

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TABLE IV

PER-CLASS QUANTITATIVE COMPARISON ON SECOND. PERCENTAGE METRICS (%) ARE REPORTED FOR $F1^c$, IoU^c, $mF1^c$, AND $mIoU^c$. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY

Method	Water		Ground		Low Veg.		Tree		Building		Playground		Average	
	$F1^c$	IoU^c	$F1^c$	IoU^c	$F1^c$	IoU^c	$F1^c$	IoU^c	$F1^c$	IoU^c	$F1^c$	IoU^c	$mF1^c$	$mIoU^c$
Foundation-model-based training-free CD														
SCM [52]	2.20	1.11	22.83	12.88	18.77	10.35	6.50	3.36	36.15	22.06	1.31	0.66	14.63	8.40
AnyChange [51]	2.19	1.11	38.90	24.15	24.82	14.17	7.49	3.89	42.79	27.22	1.03	0.52	19.54	11.84
Inst-CEG [56]	18.24	10.03	0.00	0.00	1.14	0.57	17.59	9.64	38.27	23.66	31.70	18.84	17.82	10.46
OVCD														
DynamicEarth (M-C-I) [20]	26.64	15.37	43.80	28.04	35.06	21.25	26.89	15.54	55.70	38.60	36.97	22.67	37.51	23.58
DynamicEarth (I-M-C) [20]	21.97	12.34	0.00	0.00	0.79	0.40	19.64	10.89	48.19	31.74	42.43	26.93	22.17	13.72
Seg2Change [21]	30.97	18.32	52.69	35.77	34.13	20.57	29.76	17.48	**76.01**	**61.30**	26.89	15.54	41.74	28.16
Ours	**39.71**	**24.77**	**54.14**	**37.12**	**46.82**	**30.56**	**34.46**	**20.82**	66.81	50.16	**61.07**	**43.96**	**50.50**	**34.57**

require category-aware interpretation of changed regions. On SC-SCD7, our method achieves 54.62% $mF1^c$, 38.95% $mIoU^c$, and 0.5260 mKappa, outperforming Seg2Change by 18.92, 15.72, and 0.1873, respectively. On SECOND, our method achieves 50.50% $mF1^c$, 34.57% $mIoU^c$, and 0.4678 mKappa, improving over Seg2Change by 8.76, 6.41, and 0.0750, respectively, while Seg2Change obtains a higher mOA. This difference reflects the distinct evaluation tendencies of these metrics. The mF1c and mIoUc metrics give more balanced consideration to the individual semantic change classes, whereas mOA is more strongly influenced by the dominant no-change pixels. The proposed method is designed to strengthen the discrimination of directed semantic changes rather than favoring conservative predictions of the majority no-change class. Therefore, the improvement in mF1c and mIoUc, accompanied by a lower mOA, reflects a task-oriented shift toward more balanced recognition of diverse semantic changes rather than simply a degradation in overall prediction quality. This emphasis is consistent with the objective of SCD, which requires not only localizing changed pixels but also identifying the semantic categories involved in the changes. Compared with methods that infer semantic changes from mask proposals, single-temporal semantic predictions, or endpoint comparison results, our method directly scores directed transition candidates and therefore reduces the dependence on independently inferred temporal endpoints.

2) Class-Wise Comparison on Second: Table IV provides a more detailed class-wise comparison on SECOND. The proposed method achieves the best average performance, with 50.50% $mF1^c$ and 34.57% $mIoU^c$, improving over Seg2Change by 8.76 and 6.41 percentage points, respectively. It obtains the best results on water, ground, low vegetation, tree, and playground, with $F1^c$ scores of 39.71%, 54.14%, 46.82%, 34.46%, and 61.07%, respectively, and ranks second on building. Seg2Change performs best on building, indicating that foundation-model-based semantic segmentation remains strong for visually salient and spatially compact categories. Nevertheless, our method provides more balanced performance across diverse semantic classes and achieves the best average results, showing its advantage in modeling heterogeneous semantic transitions.

TABLE V

CROSS-DATASET GENERALIZATION BETWEEN LEVIR-CD AND WHU-CD WITHOUT TARGET-DOMAIN TRAINING OR FINE-TUNING. PERCENTAGE METRICS (%) ARE REPORTED FOR $F1^c$ AND IoU^c. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	LEVIR-CD → WHU-CD		WHU-CD → LEVIR-CD	
	$F1^c$	IoU^c	$F1^c$	IoU^c
DynamicEarth [20]	66.9	50.3	28.9	16.9
Ours	**76.1**	**61.4**	**47.7**	**31.4**

3) Qualitative Comparison on Second: Fig. 5 presents qualitative comparisons on SECOND. All predictions are projected into the same semantic change map space for fair comparison. DynamicEarth tends to produce fragmented or incomplete masks when mask proposals or endpoint semantics are unreliable. Seg2Change generates more complete responses in some cases but still suffers from local false positives and incomplete boundaries in complex scenes. In contrast, our method produces cleaner and more spatially consistent semantic change maps in many complex regions, especially for low vegetation, water, and playground changes. The visual results are consistent with the quantitative improvements in Tables III and IV, further verifying the effectiveness of explicit semantic transition modeling. The relatively regular regions in some predictions of our method arise mainly from the ROI-based prediction and reconstruction procedure. Each ROI is cropped and resized to a fixed resolution, and its pixels are evaluated using a shared candidate transition set. The resulting ROI-level logits are resized and written back to the corresponding image region, while pixels outside all selected ROIs remain assigned to the no-change state. Max-score fusion is used for overlapping ROIs. Consequently, the crop–resize–write-back process and the separation between ROI-covered and uncovered regions may produce locally smooth or block-like boundaries. No manual editing or additional mask regularization is applied to the visualized results. Despite the overall improvements, Fig. 5 also shows several failure cases, including slight boundary overexpansion and missed small changed regions. These errors mainly arise when the change proposals do not fully cover the true changed areas or when the correct transition receives insufficient support in the candidate set.

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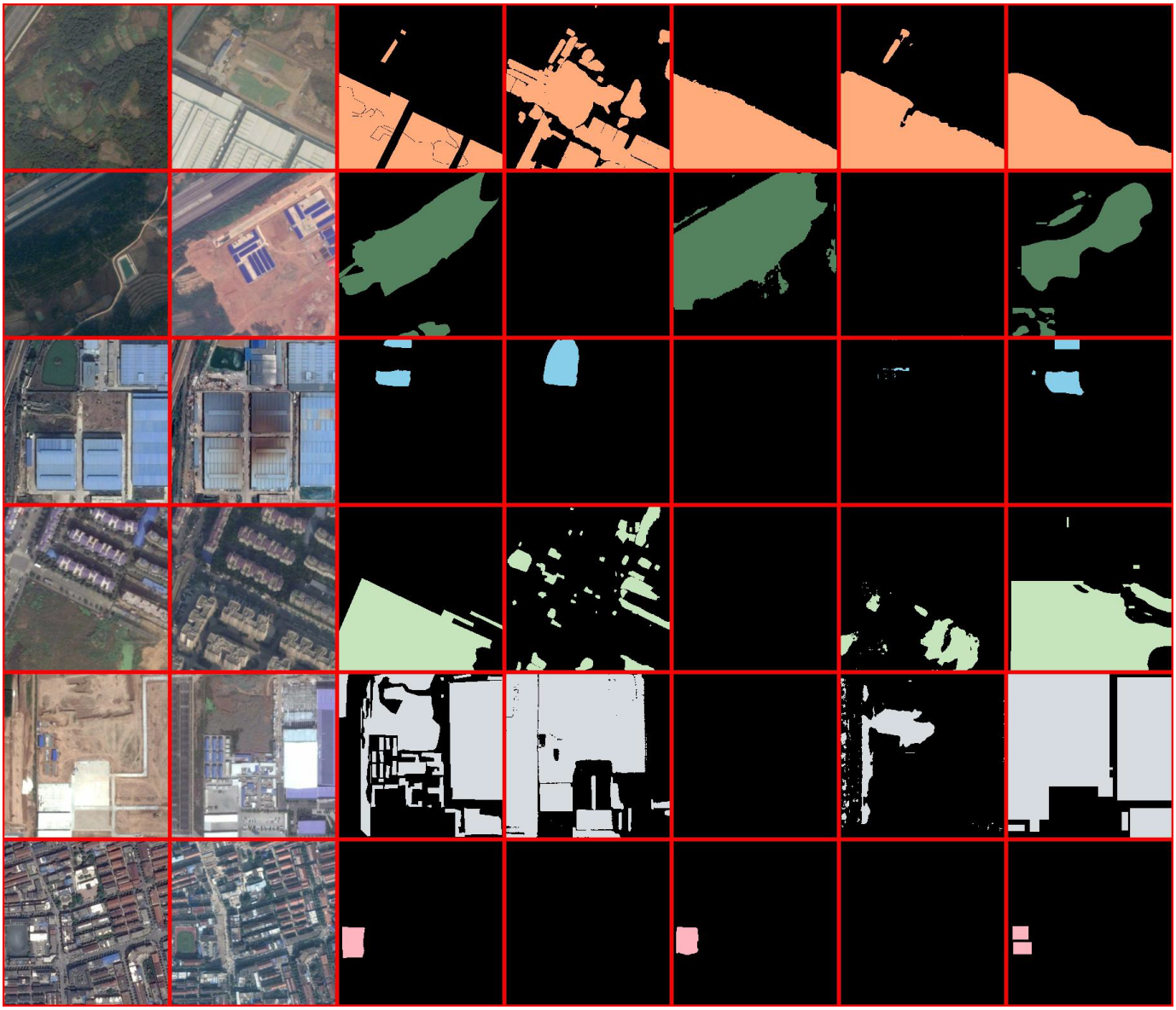


Fig. 5. Qualitative comparison on SECOND. Columns from left to right denote the prechange image, postchange image, ground truth, DynamicEarth (M-C-I), DynamicEarth (I-M-C), Seg2Change, and our method. Rows from top to bottom show representative category-wise examples of ground, tree, water, low vegetation, building, and playground changes. All predictions are projected into the same semantic change map space.

TABLE VI

ABLATION STUDY OF ENDPOINT-BASED PREDICTION, EXPLICIT TRANSITION MODELING, RCSM, AND CSTRM ON REPRESENTATIVE BENCHMARKS. THE ENDPOINT-BASED BASELINE AND THE TRANSITION-MODELING SETTING USE THE SAME FROZEN FOUNDATION REPRESENTATIONS, TRAINING DATA, AND EVALUATION PROTOCOL. PERCENTAGE METRICS (%) ARE REPORTED FOR $F1^c$, IoU^c, $mF1^c$, $mIoU^c$, AND mOA, WHILE KAPPA AND mKAPPA ARE REPORTED AS COEFFICIENTS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Setting	RCSM	CSTRM	CLCD			WHU-CD			SECOND			
			$F1^c$	IoU^c	Kappa	$F1^c$	IoU^c	Kappa	$mF1^c$	$mIoU^c$	mOA	mKappa
Baseline	–	–	71.02	55.06	0.6864	91.04	83.57	0.9071	37.69	25.43	**94.40**	**0.4907**
+ Transition modeling	–	–	75.60	60.77	0.7353	91.06	83.59	0.9074	48.40	32.68	92.38	0.4415
+ RCSM	✓	–	77.08	62.70	0.7515	91.29	83.98	0.9097	50.31	34.35	93.07	0.4642
+ CSTRM	–	✓	76.64	62.12	0.7465	91.08	83.62	0.9075	50.07	34.00	93.45	0.4638
Full Model	✓	✓	**77.14**	**62.79**	**0.7526**	**91.64**	**84.56**	**0.9133**	**50.50**	**34.57**	93.39	0.4678

4) Cross-Dataset Generalization: To further evaluate the transferability of the proposed method, we conduct bidirectional cross-dataset experiments between LEVIR-CD and WHU-CD without target-domain training or fine-tuning. As shown in Table V, our method achieves 76.1% $F1^c$ and 61.4% IoU^c on LEVIR-CD → WHU-CD, outperforming the results reported in DynamicEarth [20] by 9.2 and 11.1 percentage points, respectively. For WHU-CD → LEVIR-CD, our

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method achieves 47.7% $F1^c$ and 31.4% IoU^c, with improvements of 18.8 and 14.5 points. These results demonstrate stronger robustness to cross-dataset domain shifts without target-domain adaptation.

D. Ablation Study

Table VI evaluates the effectiveness of endpoint-based prediction, explicit transition modeling, RCSM, and CSTRM on three representative benchmarks, including CLCD for land-cover change detection, WHU-CD for building change detection, and SECOND for SCD. The baseline uses endpoint-based prediction, while the + Transition modeling setting replaces endpoint composition with direct transition prediction using the initial candidate transition set. The two settings use the same frozen DINOv2 and SegEarth-OV3 representations, and differ mainly in whether the final transition is obtained through endpoint composition or direct transition scoring. Compared with the endpoint-based baseline, explicit transition modeling substantially improves the change-sensitive metrics, increasing $F1^c$ from 71.02% to 75.60% on CLCD and $mF1^c$ from 37.69% to 48.40% on SECOND. Under this controlled setting, these improvements provide direct evidence that optimizing transition labels from joint bi-temporal evidence is more effective for semantic change recognition than deriving them through two independent endpoint decisions. Therefore, the observed gains are not solely attributable to the strong representations of DINOv2 and SegEarth-OV3 but also arise from the proposed explicit transition prediction formulation. Meanwhile, the endpoint-based baseline retains the highest mOA and mKappa on SECOND, with 94.40% and 0.4907, respectively. This result reflects the task-oriented preference of explicit transition modeling. SCD is characterized by a dominant no-change class and relatively sparse semantic change classes. Consequently, mOA and mKappa are strongly affected by overall pixel-level agreement, particularly the large number of no-change pixels, whereas mF1c and mIoUc place more balanced emphasis on the recognition of individual semantic change classes. Explicit transition modeling is designed to strengthen transition-level discrimination and reduce the conservative tendency toward the dominant no-change class. Therefore, the increase in mF1c and mIoUc, together with the decrease in mOA and mKappa, should be understood as a shift toward stronger and more balanced semantic change recognition rather than simply as a degradation in prediction quality.

Using the explicit transition model with the initial candidate transition set as the reference, adding RCSM further improves the primary F1/IoU-based metrics on all three datasets. Specifically, RCSM increases $F1^c$ from 75.60% to 77.08% on CLCD and from 91.06% to 91.29% on WHU-CD. On SECOND, it improves $mF1^c$ from 48.40% to 50.31% and $mIoU^c$ from 32.68% to 34.35%. These improvements show that supplementing residual transition candidates can effectively alleviate candidate omission, especially in scenarios with more complex semantic transition spaces.

CSTRM also improves the explicit transition model on CLCD and SECOND, increasing $F1^c$ from 75.60% to 76.64% on CLCD and $mF1^c$ from 48.40% to 50.07% on SECOND. This indicates that consistency-aware transition refinement can enhance the reliability of transition logits by incorporating TMC and spatial refinement. On WHU-CD, the improvement brought by CSTRM alone is marginal, which suggests that single-category building change detection is less sensitive to transition-logit refinement than multiclass semantic change scenarios. When RCSM and CSTRM are combined, the full model achieves the best performance on the primary F1/IoU-based metrics of all three datasets, reaching 77.14% $F1^c$ on CLCD, 91.64% $F1^c$ on WHU-CD, and 50.50% $mF1^c$ on SECOND. These results demonstrate that RCSM and CSTRM are complementary. RCSM improves candidate completeness, while CSTRM improves the semantic and spatial reliability of transition prediction.

TABLE VII

ABLATION STUDY OF TMC AND TSR IN CSTRM ON SECOND. PERCENTAGE METRICS (%) ARE REPORTED FOR $mF1^c$, $mIoU^c$, AND MOA, WHILE MKAPPA IS REPORTED AS A COEFFICIENT. RCSM IS DISABLED IN ALL SETTINGS, AND THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Setting	TMC	TSR	SECOND			
			$mF1^c$	$mIoU^c$	mOA	mKappa
Baseline	–	–	48.40	32.68	92.38	0.4415
+ TMC	✓	–	48.41	32.76	92.67	0.4431
+ TSR	–	✓	49.69	33.72	93.07	0.4581
CSTRM	✓	✓	**50.07**	**34.00**	**93.45**	**0.4638**

TABLE VIII

PROMPT WORDING ROBUSTNESS ON SECOND. ALL SETTINGS USE THE SAME DATASET SUPERVISION AND EVALUATION PROTOCOL, WHILE THE SEMANTIC PROMPTS ARE CHANGED

Prompt Setting	$mF1^c$	$mIoU^c$	mOA	mKappa
Name-only	48.72	32.94	93.33	0.4497
Synonym	**50.50**	**34.57**	**93.39**	**0.4678**
Description	48.13	32.36	92.75	0.4407

Table VII further analyzes the internal components of CSTRM on SECOND, with RCSM disabled in all settings. Compared with the baseline, TMC alone slightly improves $mF1^c$ from 48.40% to 48.41% and $mIoU^c$ from 32.68% to 32.76%, indicating that TMC provides a modest calibration effect when used independently. TSR contributes a larger improvement, increasing $mF1^c$ to 49.69% and $mIoU^c$ to 33.72%. This shows that TSR is important for improving local consistency and suppressing noisy transition responses.

Compared with the baseline, TMC alone provides only a modest improvement in the final change detection metrics, whereas TSR contributes a larger standalone gain. This difference is consistent with their distinct roles. TMC performs transition-marginal-based semantic calibration without spatial interaction, while TSR explicitly refines the spatial distribution of transition responses. Moreover, combining TMC with TSR further improves $mF1^c$ from 49.69% to 50.07% and $mIoU^c$ from 33.72% to 34.00%, indicating that the two components provide complementary semantic and spatial refinement.

E. Sensitivity Analysis

1) Impact of Prompt Wording: To evaluate the robustness of the proposed method to different semantic prompt formulations, we conduct a prompt wording sensitivity analysis on

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TABLE IX

SENSITIVITY ANALYSIS OF THE TOP-k CANDIDATE CONSTRUCTION STRATEGY. PERCENTAGE METRICS (%) ARE REPORTED FOR F1^c, IOU^c, MF1^c, MIOU^c, AND MOA, WHILE KAPPA AND MKAPPA ARE REPORTED AS COEFFICIENTS. AVGC DENOTES THE AVERAGE CANDIDATE SET SIZE. RCSM AND CSTRM ARE DISABLED IN ALL SETTINGS, AND THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Setting	CLCD				WHU-CD				SECOND				
	F1^c	IoU^c	Kappa	AvgC	F1^c	IoU^c	Kappa	AvgC	mF1^c	mIoU^c	mOA	mKappa	AvgC
Top-3	73.42	58.00	0.7100	7.71	90.96	83.42	0.9062	5.76	44.35	29.44	**93.48**	0.4068	7.70
Top-4	72.17	56.46	0.6961	13.88	**91.06**	**83.59**	**0.9074**	7.71	47.19	31.67	92.72	0.4312	13.59
Top-5	73.84	58.52	0.7158	21.73	90.83	83.21	0.9049	9.66	48.40	32.68	92.38	0.4415	21.46
Top-6	**75.60**	**60.77**	**0.7353**	31.52	90.89	83.30	0.9055	11.53	**48.67**	**32.92**	93.09	**0.4480**	31.00

TABLE X

SENSITIVITY ANALYSIS OF THE TOP-m RESIDUAL TRANSITION SUPPLEMENTATION IN RCSM. PERCENTAGE METRICS (%) ARE REPORTED FOR F1^c, IOU^c, MF1^c, MIOU^c, AND MOA, WHILE KAPPA AND MKAPPA ARE REPORTED AS COEFFICIENTS. CLCD, WHU-CD, AND SECOND USE TOP-6, TOP-4, AND TOP-5 BASE CANDIDATE SETTINGS, RESPECTIVELY. CSTRM IS DISABLED IN ALL SETTINGS, AND THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Setting	CLCD			WHU-CD			SECOND			
	F1^c	IoU^c	Kappa	F1^c	IoU^c	Kappa	mF1^c	mIoU^c	mOA	mKappa
w/o RCSM	75.60	60.77	0.7353	91.06	83.59	0.9074	48.40	32.68	92.38	0.4415
$m = 2$	76.05	61.35	0.7398	91.14	83.72	0.9082	49.10	33.13	92.89	0.4512
$m = 4$	**77.08**	**62.70**	**0.7515**	90.85	83.24	0.9051	49.10	33.34	**93.42**	0.4540
$m = 6$	74.99	59.99	0.7267	**91.29**	**83.98**	**0.9097**	49.28	33.53	93.07	0.4540
$m = 8$	75.41	60.53	0.7323	90.85	83.24	0.9052	**50.31**	**34.35**	93.07	**0.4642**

SECOND. As shown in Table VIII, all settings follow the same supervision and evaluation protocol, while only the semantic prompts are changed. The name-only setting uses only category names as prompts, the synonym setting expands each category with several semantically related expressions, and the description setting uses more descriptive textual prompts.

The results show that the proposed method is relatively stable under different prompt formulations. The synonym-based prompts achieve the best performance, with 50.50 mF1^c, 34.57 mIoU^c, 93.39 mOA, and 0.4678 mKappa. Compared with the name-only setting, synonym prompts improve mF1^c and mIoU^c by 1.78 and 1.63 points, respectively, indicating that moderate synonym expansion can provide richer semantic grounding for open-vocabulary transition recognition. In contrast, the description-based prompts lead to slightly lower performance than the name-only setting. This suggests that overly descriptive prompts may introduce redundant or ambiguous textual cues, which can weaken the alignment between visual regions and semantic transition prototypes.

2) Impact of Top-k Candidate Construction: Table IX analyzes the influence of the Top-k candidate construction strategy, with RCSM and CSTRM disabled in all settings. Increasing k enlarges the candidate transition set and reduces the risk of omitting plausible semantic transitions, as reflected by the steady increase of AvgC on all three benchmarks. However, the performance does not increase monotonically with the candidate size. On CLCD, Top-6 achieves the best result, improving F1^c from 73.42% with Top-3 to 75.60%. On WHU-CD, Top-4 obtains the best performance with 91.06% F1^c, 83.59% IoU^c, and 0.9074 Kappa, while further increasing k brings no additional gain. On SECOND, Top-6 provides the best mF1^c, mIoU^c, and mKappa, but Top-3 obtains the highest mOA. These results indicate that simply enlarging the candidate set is not always beneficial, because additional endpoint combinations may also introduce irrelevant transition competitors. Therefore, a proper candidate size should balance candidate coverage and transition discrimination.

3) Impact of Top-m Residual Supplementation: Table X evaluates the number of residual transitions supplemented by RCSM. Different datasets prefer different values of m, reflecting their different transition-space complexity and category ambiguity. On CLCD, m = 4 achieves the best performance, increasing F1^c from 75.60% without RCSM to 77.08%. On WHU-CD, m = 6 performs best, reaching 91.29% F1^c, 83.98% IoU^c, and 0.9097 Kappa. On SECOND, although Top-6 gives the best SECOND baseline when RCSM/CSTRM are disabled, we use Top-5 in the full model to avoid an overly large candidate set after residual supplementation. The best mF1^c, mIoU^c, and mKappa are obtained when m = 8, improving mF1^c from 48.40% to 50.31% and mIoU^c from 32.68% to 34.35%. These results show that residual candidate supplementation is effective, but the optimal supplementation size is dataset-dependent. A small m may be insufficient to recover omitted transitions, whereas an overly large m may introduce noisy candidates. Overall, RCSM provides a controllable way to improve candidate completeness without simply replacing the initial Top-k candidate construction.

F. Efficiency Analysis

Table XI compares the model efficiency of different OVCD methods on SC-SCD7. The proposed method achieves the highest SCD accuracy, improving mF1^c from 35.70% of Seg2Change to 54.62%. In the noncached setting, it reduces the GPU memory usage from 6.08 GB of Seg2Change to 5.19 GB, while the inference time increases to 11 528 ms/sample. This is mainly caused by the overhead of loading and running the frozen foundation models online, including DINOv2 and SegEarth-OV3 inference

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TABLE XI

EFFICIENCY COMPARISON ON SC-SCD7. PERCENTAGE METRICS (%) ARE REPORTED FOR mF1^c. "CACHE" INDICATES WHETHER PRECOMPUTED FROZEN FOUNDATION REPRESENTATIONS ARE REUSED DURING INFERENCE. THE CACHED RESULT IS REPORTED FOR REFERENCE AND IS NOT INCLUDED IN THE RANKING. THE BEST AND SECOND-BEST VALID RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY

Method	Cache	SC-SCD7 mF1^c (%) ↑	GPU Memory Usage (GB) ↓	Inference Time (ms/sample) ↓	Learnable Params (M) ↓
DynamicEarth (M-C-I) [20]	×	29.11	7.33	<u>5035</u>	/
DynamicEarth (I-M-C) [20]	×	9.19	15.33	6784	/
Seg2Change [21]	×	<u>35.70</u>	<u>6.08</u>	**1521**	**3.90**
Ours	×	**54.62**	**5.19**	11528	<u>5.84</u>
	✓	54.62	0.82	403	5.84

for full feature extraction. When the precomputed frozen representations are reused, the online stage reduces to the lightweight proposal and transition prediction modules, and the memory usage and inference time decrease to 0.82 GB and 403 ms/sample, respectively. This cached result should therefore be interpreted with methods under the same cache setting rather than with methods that extract foundation features online.

In terms of learnable parameters, our method contains 5.84M trainable parameters, which is slightly larger than Seg2Change but remains lightweight compared with large foundation-model backbones. The noncached setting reflects an end-to-end deployment where frozen foundation representations are extracted online, whereas the cached setting matches the training and evaluation protocol in which these frozen representations are precomputed. Under this cached protocol, only the lightweight proposal and transition prediction modules are executed online. Overall, the proposed method achieves strong SCD accuracy, with efficiency depending on whether frozen representations are cached.

We further analyze scalability with respect to the vocabulary size. For a vocabulary of N semantic categories, the complete directed transition space contains $N(N-1)+1$ states. However, explicit pixel-level transition scoring is performed only over the compact candidate set. Since the initial set contains at most k^2+1 transitions and RCSM supplements at most m residual transitions, the final candidate size satisfies $|\mathcal{C}| \le k^2+m+1$. For example, under the SECOND setting ($k = 5$, $m = 8$), at most 34 transitions enter pixel-level scoring. When $N = 100$, 200, and 500, the corresponding complete transition spaces contain 9901, 39 801, and 249 501 states, respectively, while the pixel-level candidate size remains bounded by 34. Note that RCSM still performs ROI-level residual scoring over the global transition space; thus, the candidate mechanism primarily avoids quadratic growth in the more expensive pixel-level transition scoring stage.

V. CONCLUSION

This article presents an explicit semantic transition modeling framework for OVCD, which directly represents changed pixels as explicit transitions instead of composing two independently predicted endpoint labels. With frozen DINOv2 and SegEarth-OV3 representations, the proposed method performs proposal-guided sparse transition scoring, supplements omitted candidates through RCSM, and refines transition logits through CSTRM. Experiments on six benchmarks demonstrate consistent improvements over representative baselines, and ablation studies verify the complementary effects of residual candidate supplementation and consistency-aware refinement. Future work focuses on more efficient online foundation feature extraction and weaker supervision for open-vocabulary transition learning, and dedicated category-disjoint or transition-disjoint evaluation protocols.

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